Written Homework Assignment #1

Problem 1.1 (Stephens, page 13)

Basic Tasks in Software Engineering Projects:

1. Requirements Gathering
2. System Design
3. Implementation (Coding)
4. Testing
5. Deployment
6. Maintenance

Problem 1.2 (Stephens, page 13)

One Sentence Descriptions of the Tasks:

1. Requirements Gathering – Identify and document stakeholder needs and expectations.
2. System Design – Define architecture, components, interfaces, and data models.
3. Implementation – Convert design specifications into working software through coding.
4. Testing – Validate functionality, performance, and security by running various test cases.
5. Deployment – Release software for end-users, ensuring it operates in the target environment.
6. Maintenance – Provide updates, bug fixes, and improvements after deployment.

Problem 2.4 (Stephens, page 27)

Google Docs Versioning vs. GitHub Versions:

- Google Docs provides a version history feature, allowing users to track document changes and revert to previous versions. GitHub uses commit history, where changes are recorded with commit messages, and users can branch, merge, and collaborate effectively.
- Differences: Google Docs is limited to textual changes, while GitHub supports a broad range of software development workflows.
- Similarities: Both platforms maintain historical records of changes and allow reverting to previous states.

Problem 2.5 (Stephens, page 27)

JBGE Meaning:

- JBGE (Just Barely Good Enough) refers to the principle of delivering a product that meets minimum acceptable standards without unnecessary complexity or perfectionism.

Problem 4.2 (Stephens, page 78)

Critical Path Analysis:

1. Calculate total expected time for each task.
2. Identify the longest path in the project schedule.
3. The critical path consists of tasks: A → D → E → M → Q.
4. Total Project Duration: 29 working days.

Problem 4.4 (Stephens, page 78)

Gantt Chart Creation:

- Start date: Wednesday, January 1, 2024.
- Work schedule: Exclude weekends and holidays.
- Plot task dependencies along the timeline.

Problem 4.6 (Stephens, page 79)

Handling Unpredictable Problems:

- Risk Management – Identify potential risks early.
- Contingency Planning – Prepare alternate plans for disruptions.
- Agile Methodologies – Adapt quickly to changes.
- Buffer Time – Include extra time in schedules for unforeseen issues.
- Effective Communication – Keep stakeholders informed of risks and solutions.

Problem 4.8 (Stephens, page 79)

Biggest Mistakes in Task Tracking:

1. Ignoring Task Dependencies – Leads to unrealistic schedules.
2. Lack of Regular Updates – Failing to track progress causes project derailment.

Problem 5.1 (Stephens, page 114)

Five Characteristics of Good Requirements:

1. Clear and Unambiguous – Easily understood by all stakeholders.
2. Testable – Can be verified through testing.
3. Feasible – Achievable within technical and budget constraints.
4. Relevant – Directly tied to project goals.
5. Traceable – Can be tracked throughout development.

Problem 5.3 (Stephens, page 114)

Audience-Oriented Categories for TimeShifter Requirements:

- End-User Needs – a, b, c, d, e, f, h, i, j, k, l, m, n.
- Performance and Security – g, o, p.
- System Administration – o, p.
- Are all categories covered? Yes, most areas are well-represented, though additional security measures may be considered.

Problem 5.9 (Stephens, page 115)

Brainstorming Mr. Bones Hangman Game Using the MoSCoW Method:

- Must-Have:
    - Improved user feedback messages.
    - Mobile-friendly UI enhancements.
    - More varied word selection.
- Should-Have:
    - Multiple difficulty levels.
    - Sound effects for user actions.
- Could-Have:
    - Customizable themes and backgrounds.
    - Multiplayer mode for competitive play.
- Won't-Have (for now):
    - 3D character animations.
    - Augmented reality (AR) integration.